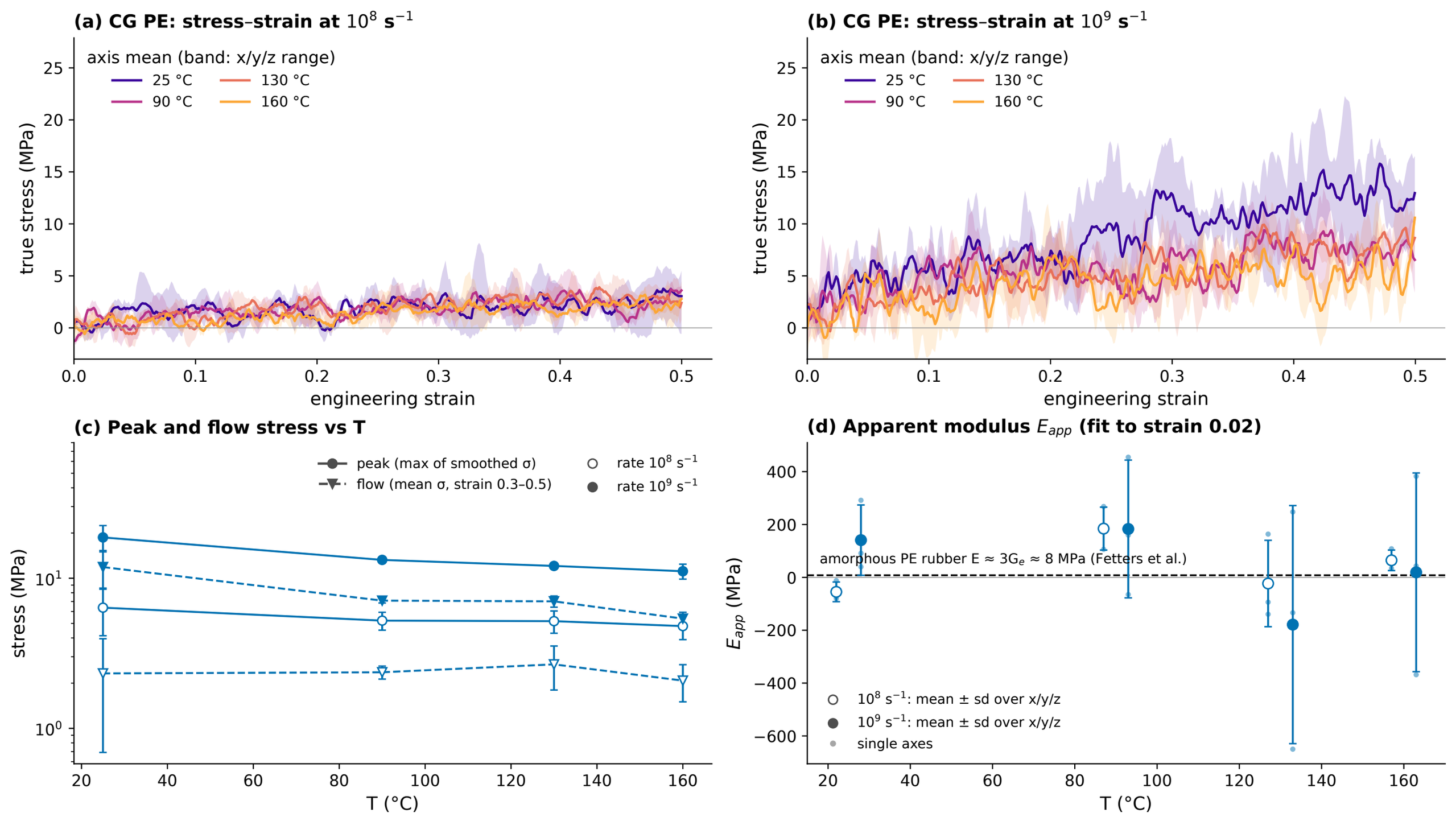

(a) CG PE: stress–strain at 10^8 s^{-1}
axis mean (band: x/y/z range)
25 °C
130 °C
90 °C
160 °C
true stress (MPa)
engineering strain
(b) CG PE: stress–strain at 10^9 s^{-1}
axis mean (band: x/y/z range)
25 °C
130 °C
90 °C
160 °C
true stress (MPa)
engineering strain
(c) Peak and flow stress vs T
peak (max of smoothed σ)
flow (mean σ, strain 0.3–0.5)
rate 10^8 s^{-1}
rate 10^9 s^{-1}
stress (MPa)
T (°C)
(d) Apparent modulus E_{app} (fit to strain 0.02)
amorphous PE rubber E ≈ 3G_e ≈ 8 MPa (Fetters et al.)
10^8 s^{-1}: mean ± sd over x/y/z
10^9 s^{-1}: mean ± sd over x/y/z
single axes
E_{app} (MPa)
T (°C)